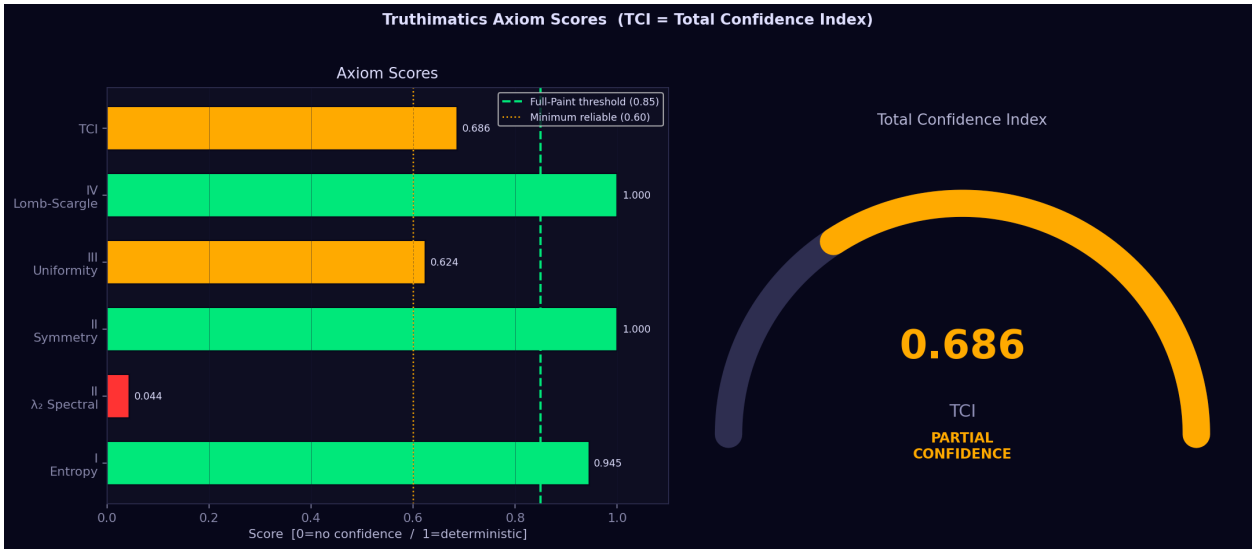
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_EA521A3362
Taxa	Giganotosaurus
Clade	Theropod
PBDB Records	4
Mass (regression)	6829.3 kg (Campione & Evans 2012)
Mass (volumetric)	5702.3 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	6.709 m³
Validity Score	1.0000 / 1.0000
TCI	0.6858 (partial)
Axiom I Entropy	0.9445
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.6235
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.588, 1.315, 0.0]
Generated	2026-04-30 23:52 UTC

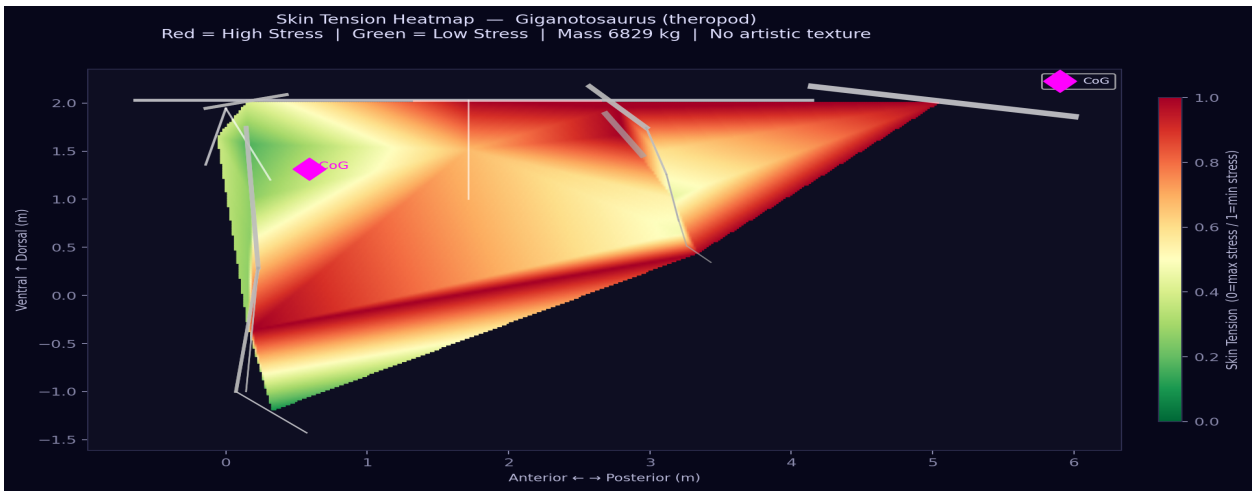
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



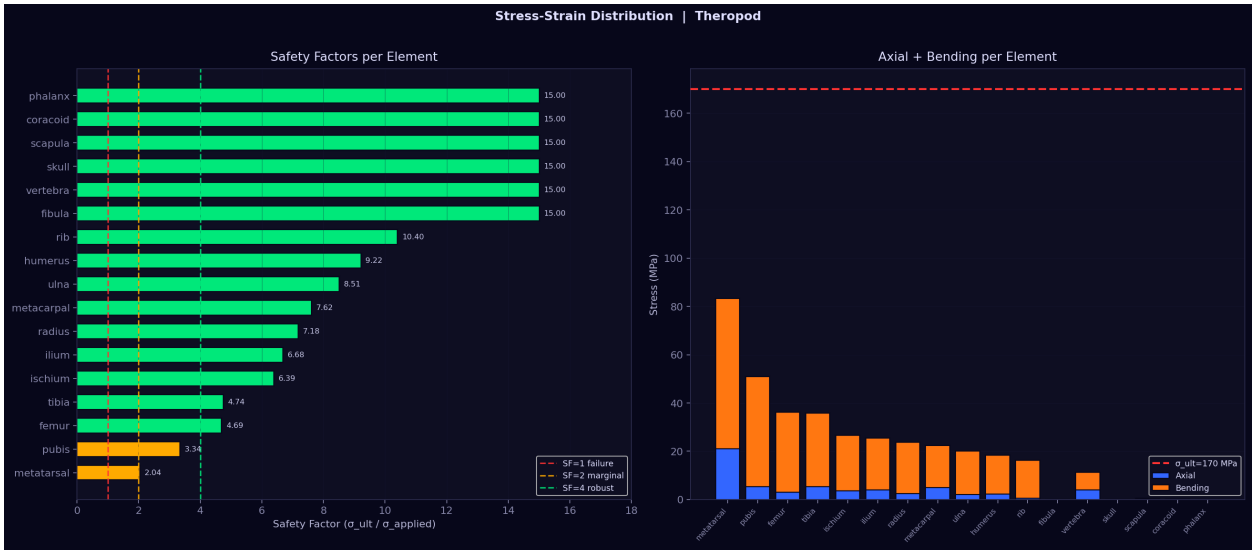
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

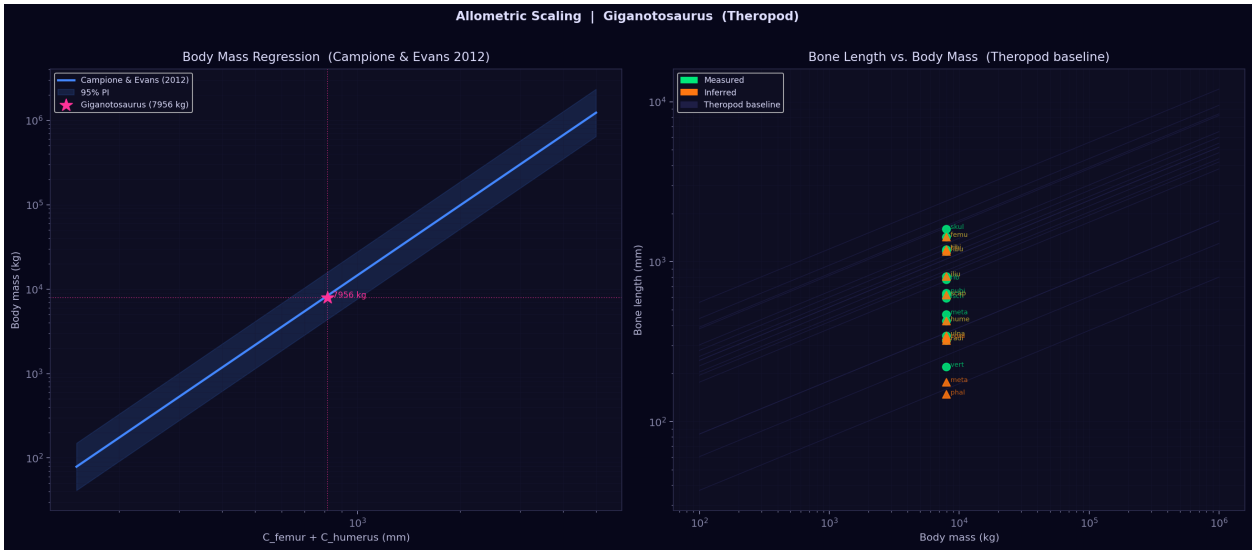


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.079	0.135	0.215	15.000	OK
femur	3.075	33.160	36.235	4.692	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	2.318	16.129	18.447	9.216	OK
ilium	3.903	21.538	25.441	6.682	OK
ischium	3.551	23.032	26.583	6.395	OK
metacarpal	4.885	17.433	22.318	7.617	OK
metatarsal	21.070	62.288	83.358	2.039	WARNING
phalanx	0.000	0.000	0.000	15.000	OK
pubis	5.328	45.629	50.957	3.336	WARNING
radius	2.368	21.323	23.691	7.176	OK
rib	0.507	15.836	16.344	10.401	OK
scapula	0.105	0.382	0.487	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	5.346	30.489	35.835	4.744	OK
ulna	2.018	17.969	19.987	8.506	OK
vertebra	3.903	7.294	11.197	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	3.093 m	Femur + tibia + metatarsal
Estimated Speed	21.0 km/h (5.84 m/s)	Gait: Trot | Fr=1.126
Stride Length	7.35 m	Alexander (1976) / Thulborn (1990)
Bite Force	64028.5 N (6526.9 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	1600.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.3911	Circumference / length (dimensionless)
Limb Strength Index	5.4e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	11.9 m	Allometric from femur length
Est. Neck Length	1.67 m	Allometric from femur length
Est. Tail Length	4.52 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	333.0	262.2	9169.55	11446.4	Constraint	0.40	15.00
femur	1430.0	178.0	18148.31	22654.5	Measured	0.88	4.69
femur_right	1430.0	178.0	18148.31	22654.5	Symmetry	0.84	4.69
fibula	1162.0	50.0	821.37	861.5	Measured	0.88	15.00
fibula_right	1162.0	50.0	821.37	861.5	Symmetry	0.84	15.00
humerus	425.0	82.0	1144.66	1428.9	Measured	0.88	9.22
humerus_right	425.0	82.0	1144.66	1428.9	Symmetry	0.84	9.22
ilium	812.0	158.0	8119.52	10135.6	Measured	0.88	6.68
ilium_right	812.0	158.0	8119.52	10135.6	Symmetry	0.84	6.68
ischium	592.0	98.0	2277.37	2842.8	Measured	0.88	6.39
metacarpal	176.8	54.8	79.11	65.1	Constraint	0.65	7.62
metatarsal	468.0	68.0	866.81	1082.0	Measured	0.88	2.04
phalanx	148.0	54.6	53.27	41.2	Constraint	0.40	15.00
pubis	638.0	80.0	1635.54	2041.6	Measured	0.88	3.34
radius	322.0	48.0	297.17	371.0	Measured	0.88	7.18
radius_right	322.0	48.0	297.17	371.0	Symmetry	0.84	7.18
rib	775.0	40.0	496.69	620.0	Measured	0.88	10.40
scapula	618.0	228.0	12868.21	16063.4	Measured	0.88	15.00
scapula_right	618.0	228.0	12868.21	16063.4	Symmetry	0.84	15.00
skull	1600.0	640.0	262506.46	327686.8	Measured	0.88	15.00
tibia	1195.0	135.0	8723.60	10889.7	Measured	0.88	4.74
tibia_right	1195.0	135.0	8723.60	10889.7	Symmetry	0.84	4.74
ulna	345.0	52.0	373.67	466.4	Measured	0.88	8.51
ulna_right	345.0	52.0	373.67	466.4	Symmetry	0.84	8.51
vertebra	220.0	158.0	2199.87	2746.1	Measured	0.88	15.00